



جامعة أم القرى
UMM AL-QURA UNIVERSITY

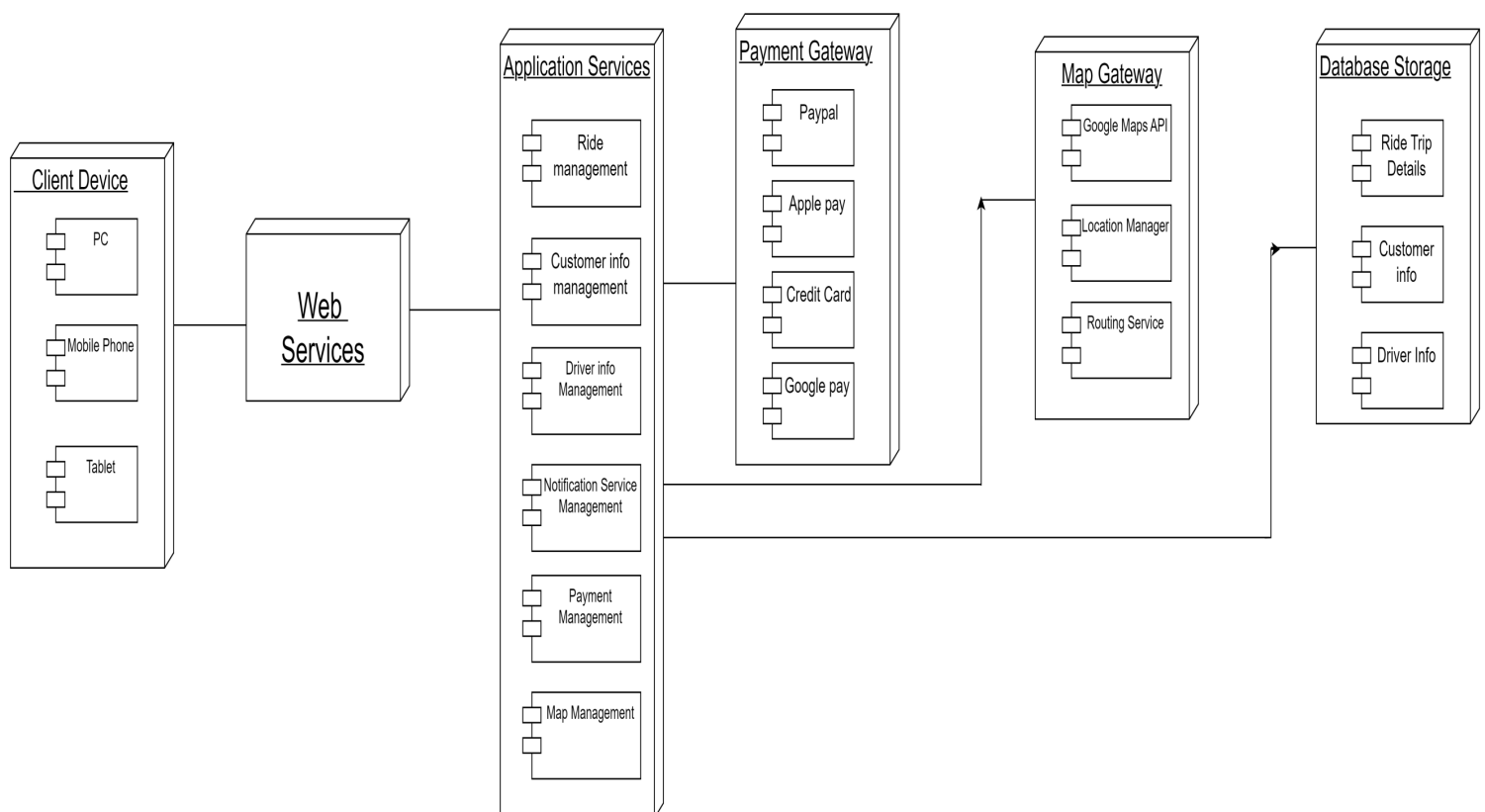
نورة محمد الكبكي 444003676 شعبة 1
اميرة محسن الغموي 444005124 شعبة 2
أمجاد احمد هوساوي 444003283 شعبة 1
وئام راشد 444015412 شعبة 1

The Ride-Booking system project

The System Overview:

In the Ride-Booking system, we seek to provide service to passengers to move to any location without the need to driving with advantages such as choosing the type of car, the number of passengers and paying in multiple ways with integrated payment management system. The system is also characterized by linking to maps, enabling users to determine their location so that the driver can reach them and deliver them to their desired point.

The deployment diagram



The Key Components of deployment diagram:

App User Interface

Our system provides user interfaces across multiple platforms:

1. **mobile phones:** such as: iOS, and android OS devices.
2. **tablets:** devices which have a different screen size.
3. **PCs:** such laptops, mac, Microsoft's PC devices.

Web Services

This node represents the internet, using secure protocols like HTTPS.

Application Services

The backend manages core application services, including:

1. **Ride Management:** Coordinates ride bookings, matches drivers with passengers, and updates ride status, driver availability, and ride details in real time.
2. **Customer Info Management:** Manages user data, including ride history.
3. **Driver Info Management:** Manages driver data, such as availability, vehicle type, ratings, and driving history.
4. **Notifications Management:** Sends real-time updates to users about ride status and notifications to drivers for ride requests.
5. **Payment Management:** Integrates with third-party payment gateways to process payments, offering multiple payment methods to passengers.
6. **Map Management:** Integrates with third-party map APIs Google Maps for driver location tracking and real-time navigation.

Payment Gateway

Handles financial transactions for the ride-booking system:

1. **Apple Pay:** Enables payments via Apple devices, ensuring secure transactions.
2. **Google Pay:** Allows Android users to pay securely.
3. **Credit Card:** Supports direct payments with credit cards like Visa and Al Rajhi Bank. Widely used by most customers.
4. **PayPal:** Provides the option to pay via PayPal, integrating with the payment gateway for seamless transactions.

Map Gateway

Integrates with the ride-booking system to provide GPS services, including driver tracking, locating passengers, and real-time navigation:

1. **Google Maps API:** Provides geolocation, map display, and directions.
2. **Location Manager:** Tracks users, drivers, and rides in real time.
3. **Routing Service:** Calculates the best routes and provides time and **distance estimates**.

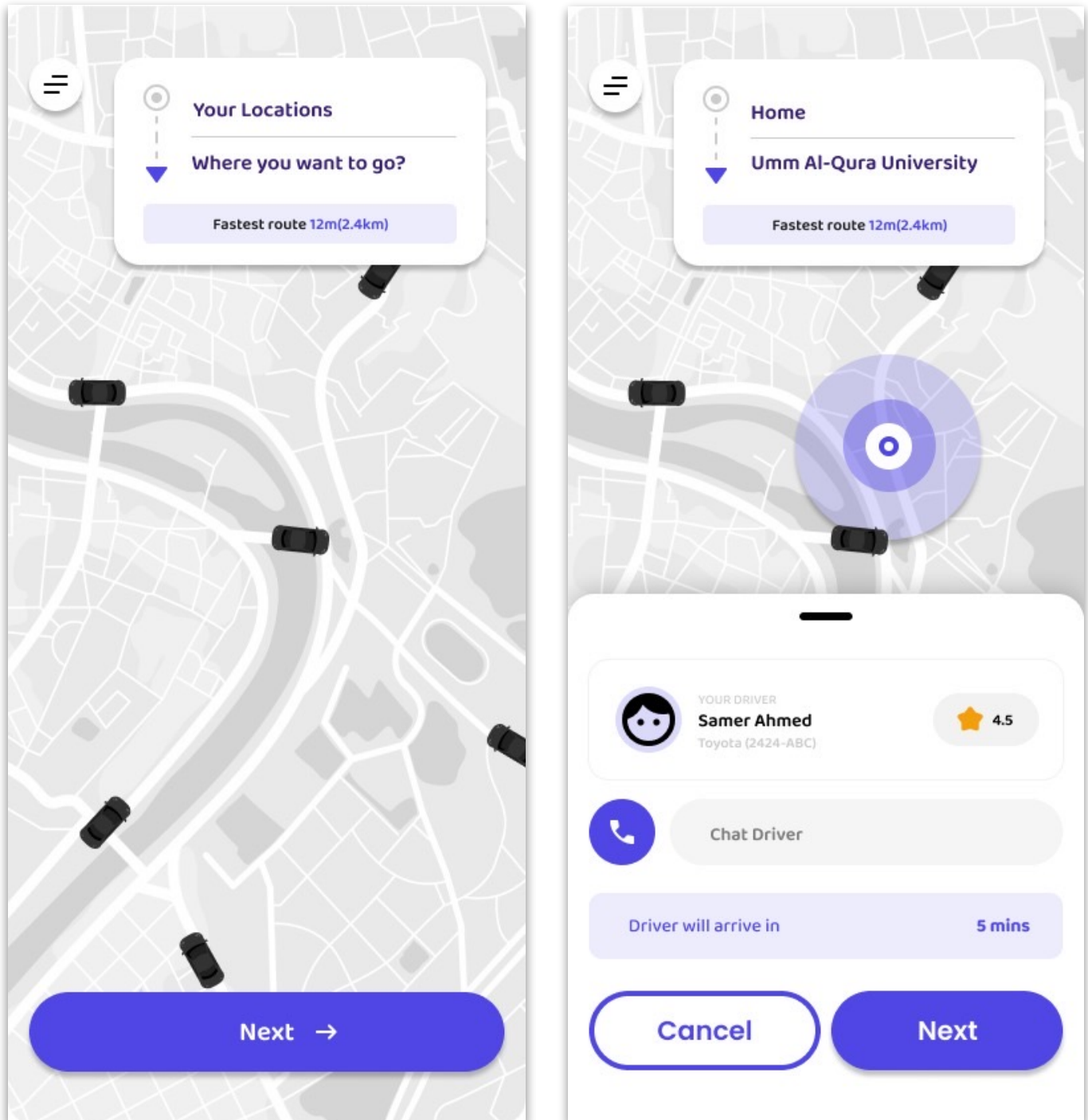
Database Storage

Supports real-time tracking and historical data storage for the Ride-Booking system including:

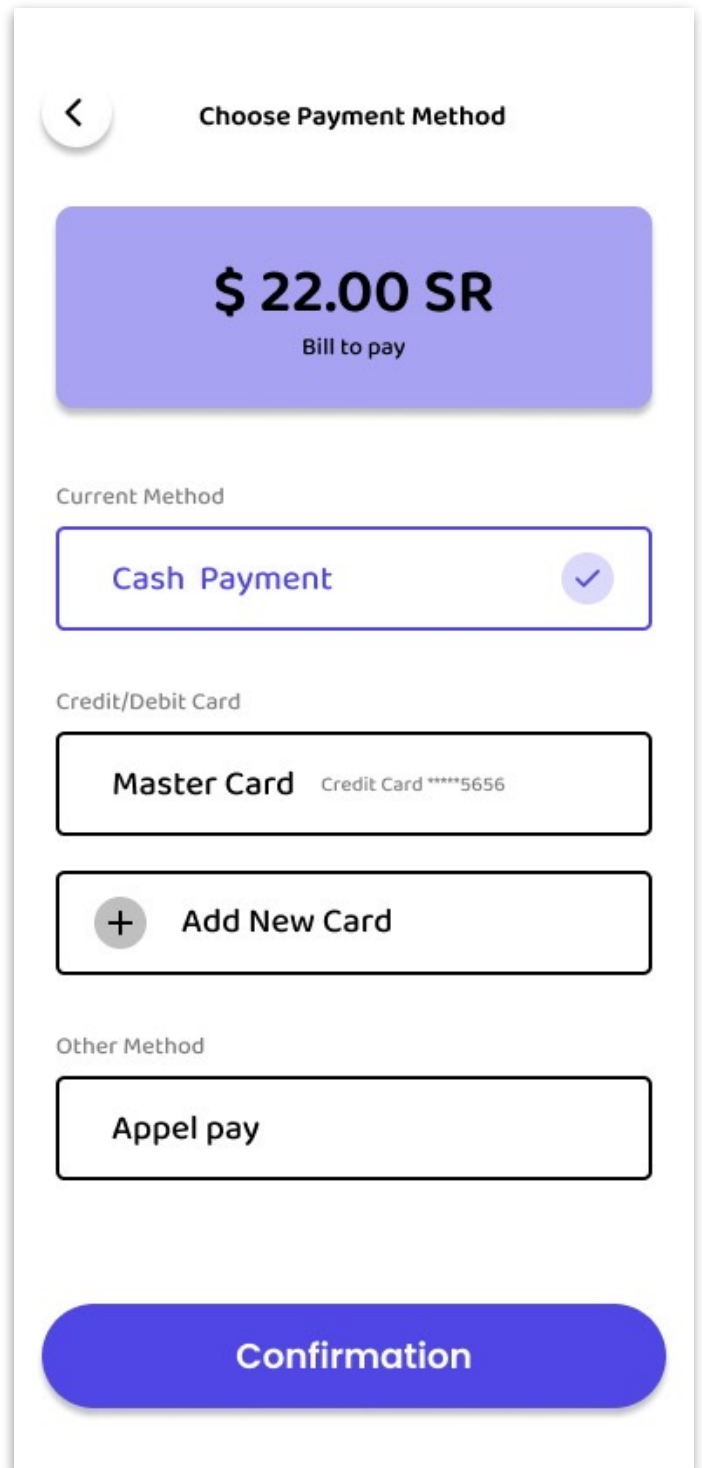
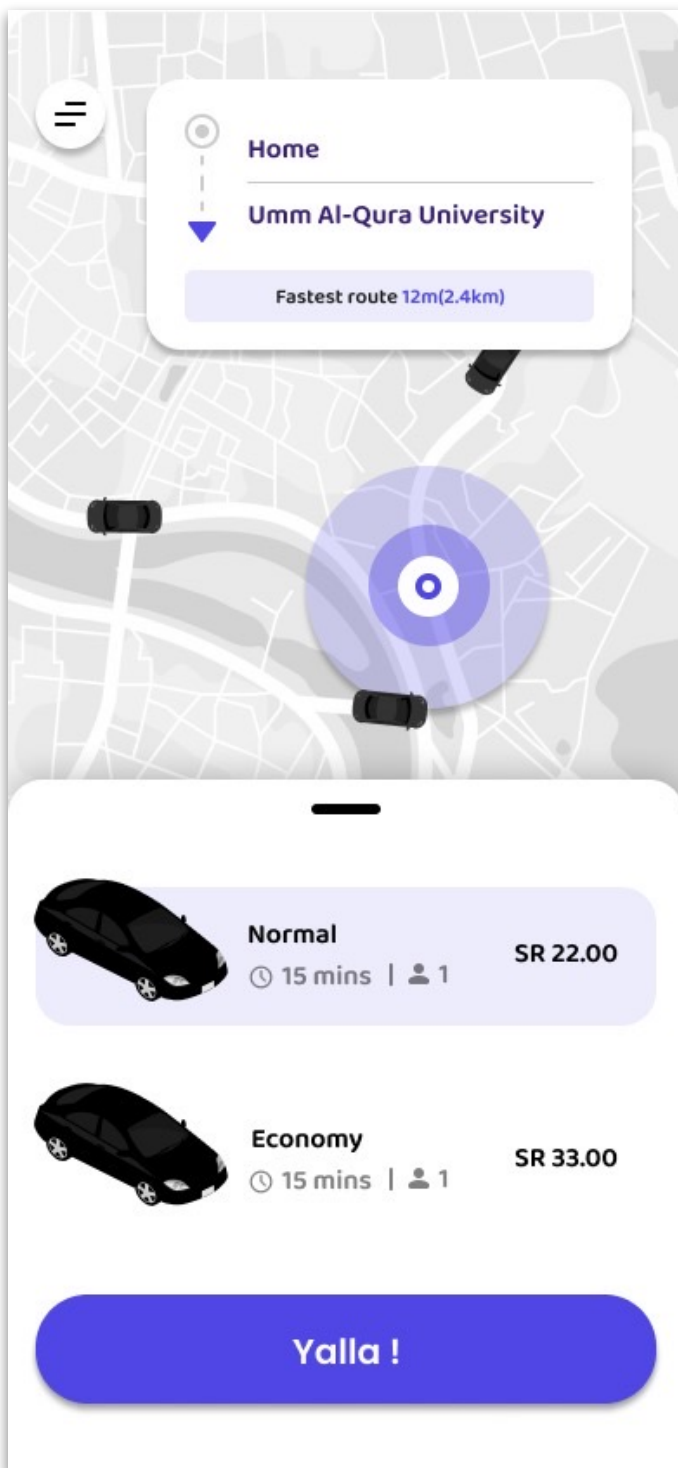
1. **Ride trip details :** Tracks ongoing and completed rides, including timestamps, routes, and fare details.
2. **Costumer Info:** Stores user profiles, ride history, and payment preferences.
3. **Driver Info:** Maintains driver profiles, availability, vehicle details, and ratings.

The figma interfaces:

<https://www.figma.com/design/tGVM9LK8ms8pxle7GJ0k7q/Untitled?node-id=174-52&t=l7Qh3CBtYKN2LBjM-1>



This the request ride interfaces.



The confirmation and payment screen.